# Bing Han, Ph.D.

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## PROFESSIONAL SUMMARY

Cancer biologist with 10+ years in colorectal cancer and 5+ years leading preclinical programs end to end — from target and mechanism through biomarker strategy to in vivo efficacy. Led a 106-analog small-molecule campaign with medicinal chemistry partners, advancing 10 compounds into efficacy studies; contributed to preclinical validation of a KRAS-targeted mRNA vaccine with a biotech collaborator. **Designs and runs the omics analysis as well as the biology** — scRNA-seq, scATAC-seq, bulk RNA-seq and ChIP-seq in R and Python. First or co-first author on 5 papers; co-author on two *Nature Communications* and two *Science Advances* papers.

## CORE TECHNICAL SKILLS

- **Drug discovery and pharmacology:** small-molecule inhibitor campaigns and SAR-guided analog triage; tiered biomarker funnels (target engagement, mechanism of action, cell killing, immune readouts); anti-PD-1 and combination efficacy studies; mRNA vaccine evaluation.
- **In vivo tumor models:** Apc-driven, AOM/DSS-driven and Kras/Apc/Trp53-mutant colorectal models; Kmt2d genetic models; syngeneic and xenograft (CDX/PDX); microbiota colonization and bone loss models. Study design and response assessment.
- **Omics and data analysis:** scRNA-seq and scATAC-seq (Seurat, Signac); bulk RNA-seq, ChIP-seq, m6A-seq, ribosome profiling; TCGA and DepMap; pathway and gene-set analysis. R and Python; reproducible, version-controlled workflows with automated testing.
- **Immune and tissue biomarkers:** flow cytometry, CyTOF immune profiling, IHC/IF, tumor immune microenvironment characterization.
- **Molecular and biochemical methods:** CRISPR gene editing, CRISPR/RNAi screening, qPCR, co-IP, Western blot, immunofluorescence.
- **Cellular systems:** intestinal organoids, cancer cell lines, primary bone marrow cultures, viability and drug sensitivity assays.
- **Disease biology:** colorectal cancer, KRAS/MAPK signaling, inflammation and microbiome-driven tumorigenesis, intestinal stemness, epigenetic and RNA regulation.

## RESEARCH EXPERIENCE

### Postdoctoral Scholar, Cancer Biology

University of California, Los Angeles | Los Angeles, CA | Apr 2021 – Sep 2026

*Key personnel on three NIH/NIDCR R01 grants (DE017684, DE019412, DE029200).*

***Small-molecule screening and in vivo validation (project lead)***

- Led a screening program for BMI1 inhibitors in colorectal and squamous cell carcinoma, working with medicinal chemistry collaborators on 106 analogs designed from the reference compound PTC596 (unesbulin).
- Built a tiered biomarker funnel so each round of analogs received a go/no-go in days rather than months: target engagement (BMI1, H2AK119ub), mechanism of action (tubulin polymerization; CDK1/2 activation via p38 MAPK), tumor cell killing, and immune chemokine readouts.
- Advanced 10 compounds into in vivo efficacy studies; **2 matched the reference compound's tumor suppression**.

***Microbiome-driven tumor immune remodeling in colorectal cancer (project lead)***

- Led a program showing that *Fusobacterium nucleatum* drives colorectal cancer through KMT2D-dependent inflammatory reprogramming of the tumor immune microenvironment (*Science Advances*, under revision).
- Established in vivo models combining microbiota exposure, Kmt2d genetic perturbation, immune profiling and therapeutic intervention; showed that targeting inflammation suppressed tumor progression and **enhanced anti-PD-1 efficacy in vivo**.
- Profiled the tumor microenvironment by flow cytometry, CyTOF and IHC/IF, and led the scRNA-seq and scATAC-seq analysis to identify myeloid programs and MDSC/macrophage enrichment; validated an IKK2–NF-κB/p65–KMT2D regulatory axis by ChIP-qPCR at inflammatory gene promoters (FOSL1, TNF).

***KRAS-targeted mRNA vaccine (industry collaboration)***

- Contributed to preclinical validation of a KRAS mutation–targeted mRNA vaccine in Kras/Apc/Trp53-mutant colorectal cancer mouse models, in collaboration with a biotech partner; supported in vivo study execution and therapeutic response assessment.

***Bone metabolism and environmental exposure (project lead)***

- Followed up a clinical observation with complementary in vitro and in vivo models to identify environmental factors influencing bone metabolism, and defined the signaling mechanisms involved in bone remodeling (manuscript in preparation).

### Visiting Scholar, Cancer Biology

Yale University | New Haven, CT | Sep 2019 – Nov 2020

- Studied epitranscriptomic regulation of pancreatic cancer metastasis and characterized cancer-associated mutations in RNA methylation enzymes (co-author, *Science Advances* 2023).

### Ph.D. Researcher, Cancer Biology

Zhejiang University | Hangzhou, China | Sep 2015 – Dec 2020

- Discovered that the m6A reader YTHDF1 is upregulated in colorectal cancer and amplifies Wnt-driven intestinal stemness through translational control (first author, *EMBO Reports* 2020).

## INDEPENDENT COMPUTATIONAL PROJECT

**KMT2D dependency across tumour lineages** · 2025 – present

- Self-directed project integrating 19 KMT2D perturbation contrasts from 14 GEO series (99 samples, 6 tissues) with DepMap CRISPR screens across 1,361 cell lines, to determine what sets a tumour cell's dependency on a chromatin regulator.
- Found the requirement is set by epithelial differentiation state rather than by lineage or mutation status, with both data types converging on the same axis independently; reported the ceiling on predictive power explicitly rather than presenting a weak signature.
- Designed the study and the statistical approach and interpreted the results; implementation was AI-assisted. The analysis is version-controlled and tested, with every step pre-registered before it was run — code and full analysis plan available on request.

## EDUCATION

**Ph.D., Cancer Biology** · Zhejiang University, Hangzhou, China · 2015–2020

**B.Med., Preventive Medicine** · Wuhan University, Wuhan, China · 2010–2015

## PROFESSIONAL SERVICE

**Review Editor** · *Frontiers in Immunology*, Inflammation section · 9 manuscripts reviewed since 2022

**Ad hoc reviewer** · *Journal of Dental Research*; *International Journal of Molecular Medicine*; *Experimental and Therapeutic Medicine*

## AWARDS

**Poster Presentation Winner** · Science Day, Division of Regenerative and Reconstructive Sciences, UCLA · 2025 and 2026

## SELECTED PUBLICATIONS

**First or co-first author**

- **Han B**, Liu W, Chen Y, et al. *Fusobacterium nucleatum* promotes colorectal cancer via KMT2D-dependent transcriptional reprogramming of tumor immune environment. *Science Advances*. Under revision.
- **Han B**, Yan S, Wei S, et al. YTHDF1-mediated translation amplifies Wnt-driven intestinal stemness. *EMBO Reports*. 2020;21:e49229.
- **Han B**, Wei S, Li F, et al. Decoding m6A mRNA methylation by reader proteins in cancer. *Cancer Letters*. 2021;518:256–265.
- **Han B**, Liu W, Chen Y, et al. Modeling *Fusobacterium nucleatum*-driven colorectal cancer in Apc-mutant mice. In: *Fusobacterium nucleatum: Methods and Protocols*.
- Zhang L*, **Han B***, Xiang J, et al. Silica nanoparticle releases SIRT6-induced epigenetic silencing of follistatin. *Int J Biochem Cell Biol*. 2018;95:27–34. (*co-first)

**Selected co-authorships**

- Liu W, Cao H, Wang J, Elmusrati A, **Han B**, et al. Histone-methyltransferase KMT2D deficiency impairs the Fanconi anemia/BRCA pathway upon glycolytic inhibition in squamous cell carcinoma. *Nature Communications*. 2024;15:6755.
- Hou J, Gao Y, **Han B**, Yan S, Wei S, Gao X. Nuclear accumulation of YTHDF1 regulates mRNA splicing in the DNA damage response. *Science Advances*. 2025;11:eado7660.
- Yan S, Zhou X, Wu C, **Han B**, et al. Adipocyte YTH N6-methyladenosine RNA-binding protein 1 protects against obesity by promoting white adipose tissue beiging in male mice. *Nature Communications*. 2023;14:1379.
- Perez-Pepe M, Desotell AW, Li H, Li W, **Han B**, et al. 7SK methylation by METTL3 promotes transcriptional activity. *Science Advances*. 2023;9:eade7500.
- Wang F, Jiang M, Zhu L, et al. (**Han B**, co-corresponding author). Lactic acid and lactylation in the progression of pulmonary fibrosis. *BIO Integration*. 2026;7(1):975.

Full publication list: **Google Scholar**